

TOPIC 2: ENVIRONMENTAL DEGRADATION – EROSION

1. Meaning of Erosion

Erosion is the process by which the topsoil or surface of the earth is worn away and carried from one place to another by agents such as water, wind, ice or human activities. It is a major form of environmental degradation that reduces soil fertility and damages land.

2. Types of Erosion

a) Water Erosion – caused by rainfall and running water. Includes:

- Splash erosion – raindrops hitting the soil
- Sheet erosion – thin layer of soil removed over a large area
- Rill erosion – small channels formed by running water
- Gully erosion – large, deep channels (common in eastern Nigeria)

b) Wind Erosion – common in dry, sandy areas; wind carries loose soil particles.

c) Coastal / Wave Erosion – waves wearing away the shoreline.

d) Human-induced Erosion – caused by deforestation, overgrazing, poor farming and construction.

3. Causes of Erosion

- Heavy rainfall and flooding
- Deforestation (cutting down trees that hold the soil)
- Overgrazing by animals
- Poor farming practices (e.g. ploughing up and down the slope)
- Construction and mining activities
- Bush burning that destroys vegetation cover
- Soft or sandy soil that is easily washed away

4. Effects of Erosion

- Loss of fertile topsoil → reduced crop yield
- Formation of gullies that destroy roads, houses and farmland
- Silting of rivers and dams
- Displacement of people and communities
- Loss of vegetation and habitats for animals
- Increased poverty and food insecurity

5. Preventive Measures

- **Terracing** – making step-like platforms on slopes to slow down water.
- **Tree planting / Afforestation** – roots hold the soil firmly.
- **Contour ridges / Contour ploughing** – ploughing across the slope, not up and down.
- **Cover cropping** – planting crops that cover the soil.
- **Mulching** – covering soil with plant materials.
- **Avoiding bush burning and overgrazing.**
- **Constructing drainage channels and check dams.**
- **Educating communities** on good land-use practices.

6. Simple Demonstration of Soil Erosion

In class or at home: place soil on a tray or board, tilt it, and pour water. Observe how soil is washed away. Compare with soil covered by grass or leaves – less soil is lost. This shows the importance of vegetation cover.

7. Summary

Erosion is the wearing away of soil by water, wind or human activity. It causes loss of farmland, gullies and poverty. Prevention includes terracing, tree planting, contour farming and protecting vegetation cover.

End of Topic 2 – Erosion